E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 15-A

Randomness

Berkeley ENGINEERING

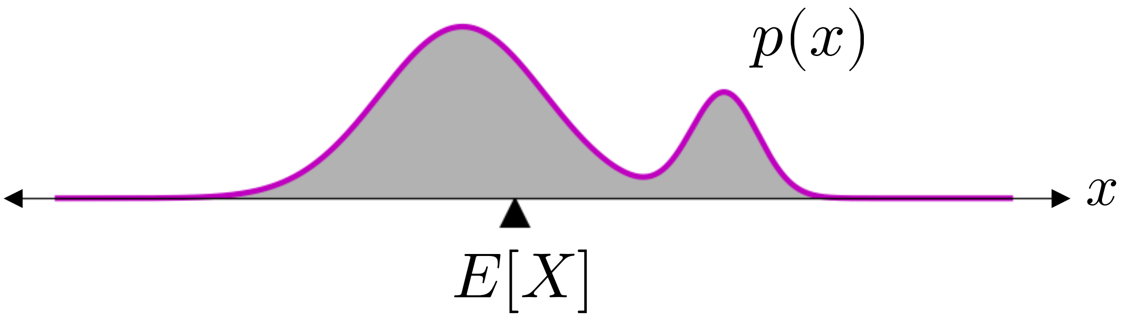
Mean of a random variable (a.k.a. *expected value*)

- X is a random variable with sample space Ω and distribution $p(x)$.
- The **mean** of X, denoted $E[X]$ or μ_X, is a real number defined with

$$E[X] = \int_{\Omega} x\, p(x)\, dx$$ ... if Ω is continuous.

$$E[X] = \sum_{x \in \Omega} x\, p(x)$$ ... if Ω is discrete.

Example: Mean of a continuous r.v.



Example: Mean of a discrete r.v.

x	$p(x)$
-1	0.2
0	0.3
2	0.5

Standard deviation of a random variable

- X is a random variable with sample space Ω and distribution $p(x)$.
- The **standard deviation** of X, denoted $\text{Std}[X]$ or σ_X, is a real number defined with

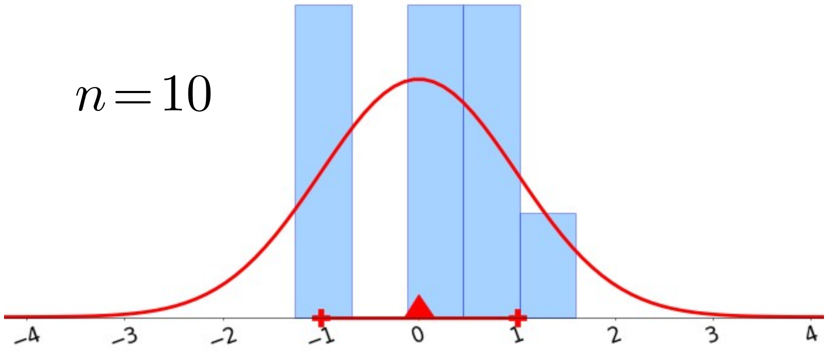
$$\text{Std}[X] = \sqrt{E[(X - \mu_X)^2]}$$

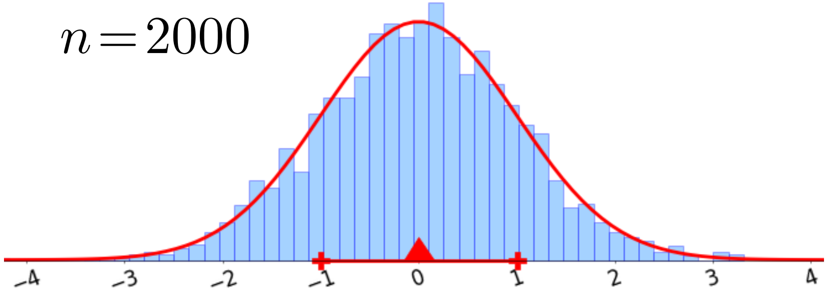
Estimating the mean and standard deviation from samples

- We have a dataset $\mathcal{D}$ consisting of n samples of X.

$$\mathcal{D} = \{x_1, \dots, x_n\}$$

- The histogram of $\mathcal{D}$ approaches the distribution of X as $n \to \infty$.





- Estimate the mean with the **sample mean** $\hat{\mu}_n$.

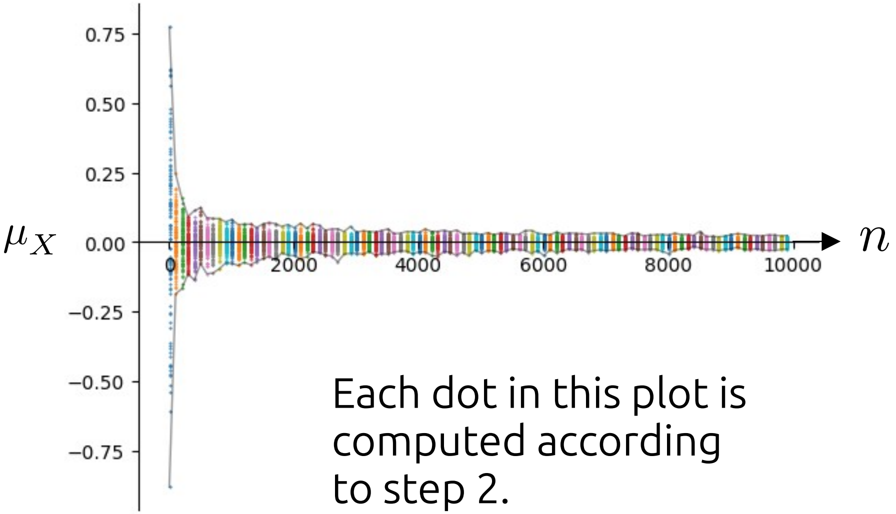
$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- Estimate the standard deviation with the **sample standard deviation** $\hat{\sigma}_n$.

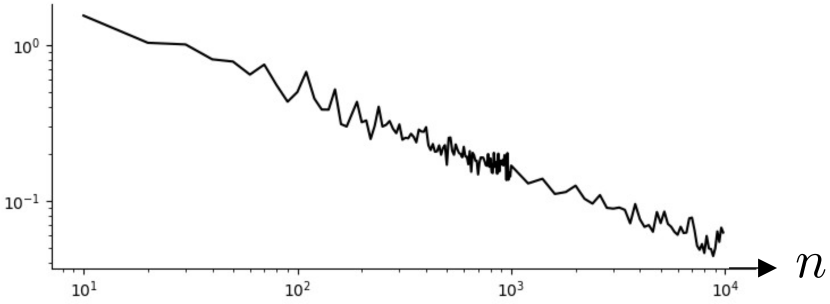
$$\hat{\sigma}_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \hat{\mu}_n)^2}$$

Convergence of the sample mean to the true mean

1. We have a random variable X with $\mu_X = 0$.
2. We collect a sample $\mathcal{D}$ with n samples and calculate $\hat{\mu}_n$.
3. Repeat this 100 times for each n.
4. Do this for increasing values of n.



Each dot in this plot is computed according to step 2.



This plot shows the variation in the 100 samples of $\hat{\mu}_n$ computed in step 3.

Justification for the "n-1" in the definition of $\hat{\sigma}_n$

- Red dots are samples of $\hat{\sigma}_n$ defined with "$n-1$".
- Blue dots are samples of $\hat{\sigma}_n$ defined with "n".
- The red and blue lines are the averages of the dots.
- The blue line underestimates the true value on average.